

Preventing Car Crashes in the City of Chicago: In-Depth Analysis and Recommendations

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Using Data Provided By The City of Chicago:
Traffic Crashes, People, and Vehicles



Executive Summary

- 1) Purpose of this assignment
- 2) Analysis process
- 3) Outcome of analysis
 - A. Summer + fall months
 - B. Traffic controls + intersections
 - C. Adverse weather + lighting
 - D. Driver errors + inexperience
 - E. O'Hare International Airport

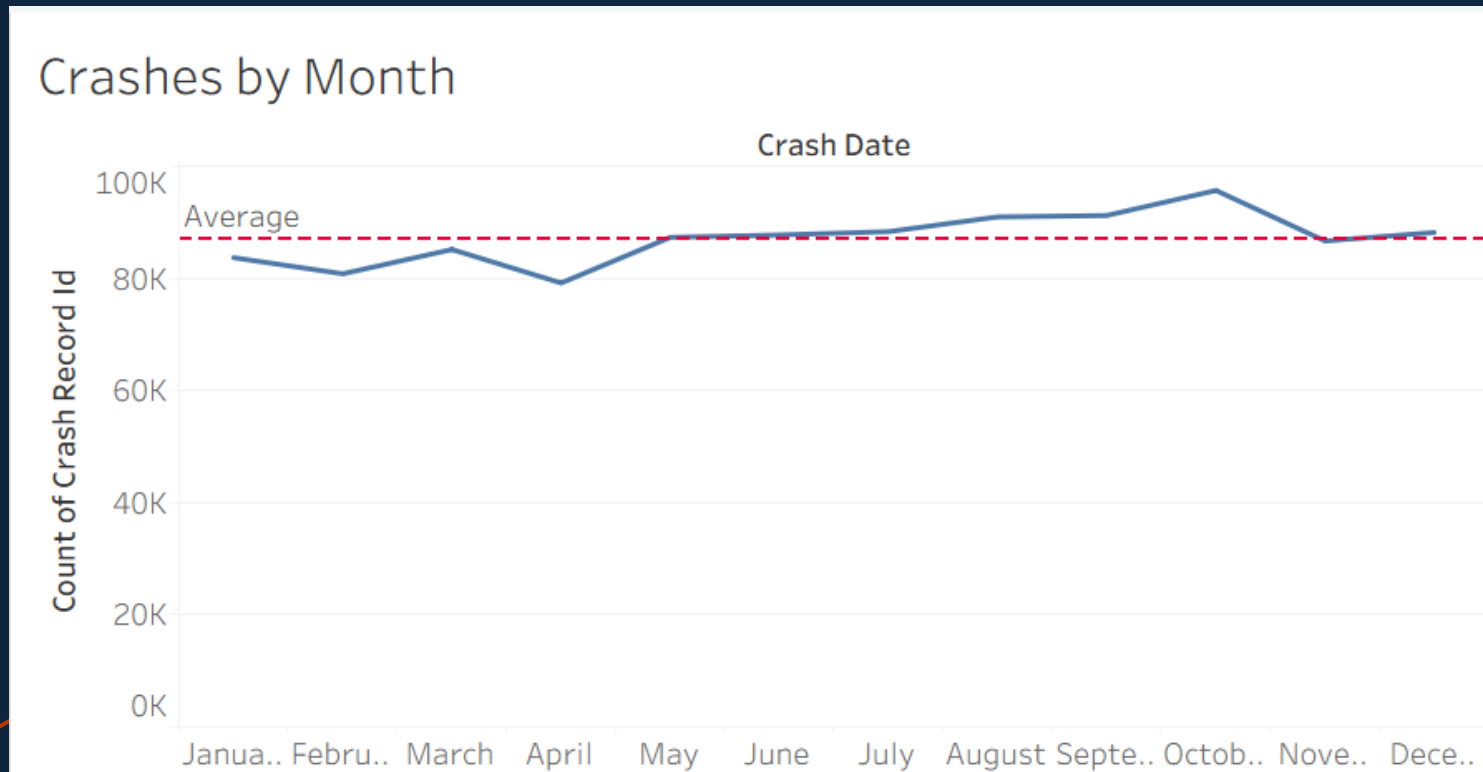




Recommendation 1: Place Speed Cameras or More Cops on Roads During the Summer and Fall

*Analysis of crash
data by month*

Recommendation 1 Visual + Analysis



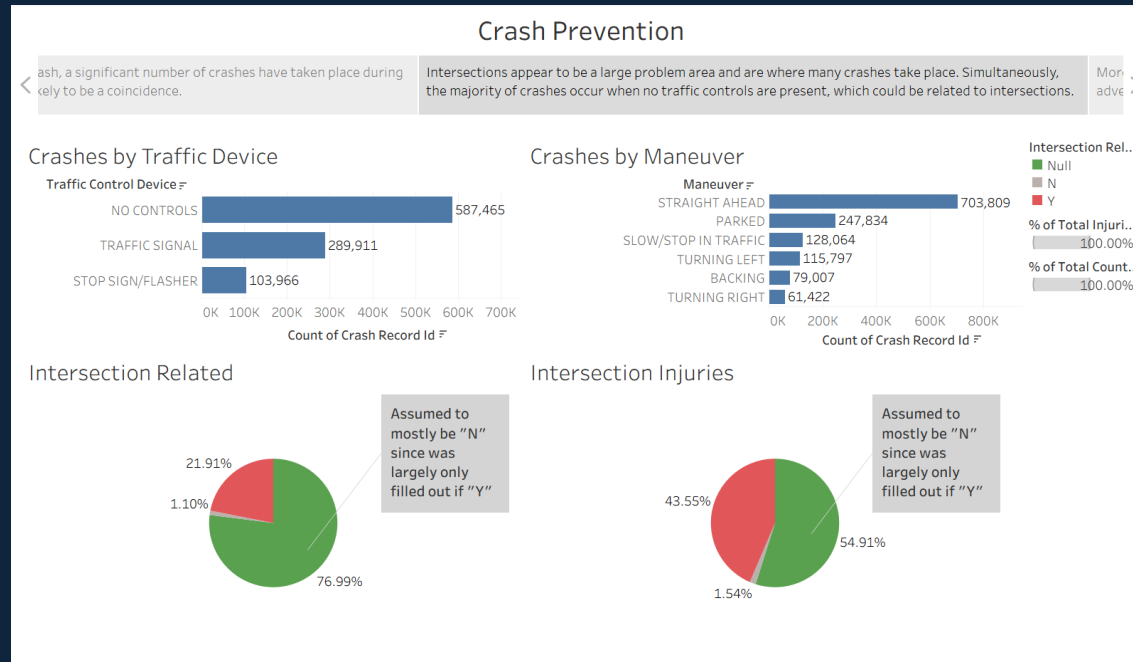
- There is a clear uptick in crashes during the summer and fall, likely due to optimal weather and more to do in the city and therefore increased tourism.
- Significantly more people in the city, especially tourists who are unfamiliar to the area and are excited about their trips, greatly increases the chances of people breaking traffic laws and getting into crashes as a result.
- If there were more officers or speed cameras in the city, drivers may be more inclined to drive more cautiously and safely, which could prevent a significant number of unnecessary crashes.



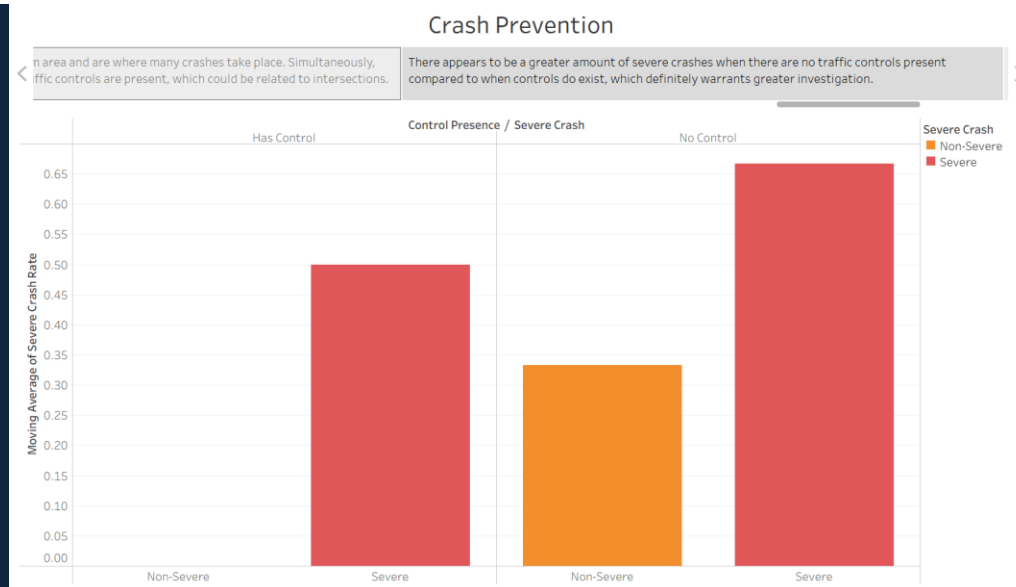
Recommendation 2: More Traffic Controls and Devices, Especially at Intersections

Analysis of crash data and injury data at intersections, crashes by traffic device, and crashes by maneuver

Recommendation 2 Visuals + Analysis



- Despite only accounting for about 22% of crashes, intersections are where nearly 44% of injuries occur.
- In addition, well over half of crashes occur where no traffic control devices are present, and most crash maneuvers match with what would occur at intersections.
- Severe crashes are also more common when there are no traffic controls present.
- Installing more traffic controls, especially at intersections, could certainly help cut down on crashes, as more direction for drivers would reduce confusion and mistakes.
- Placing cameras or more police officers at intersections could also help prevent some crashes, as people would likely be less inclined to break laws.

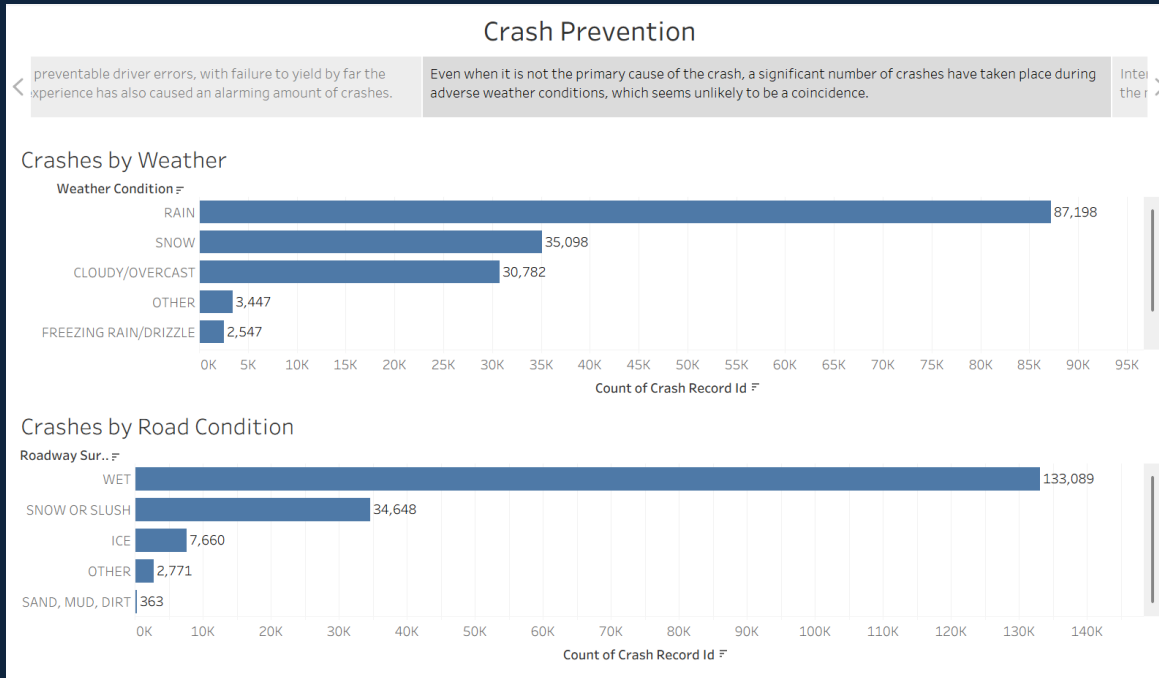




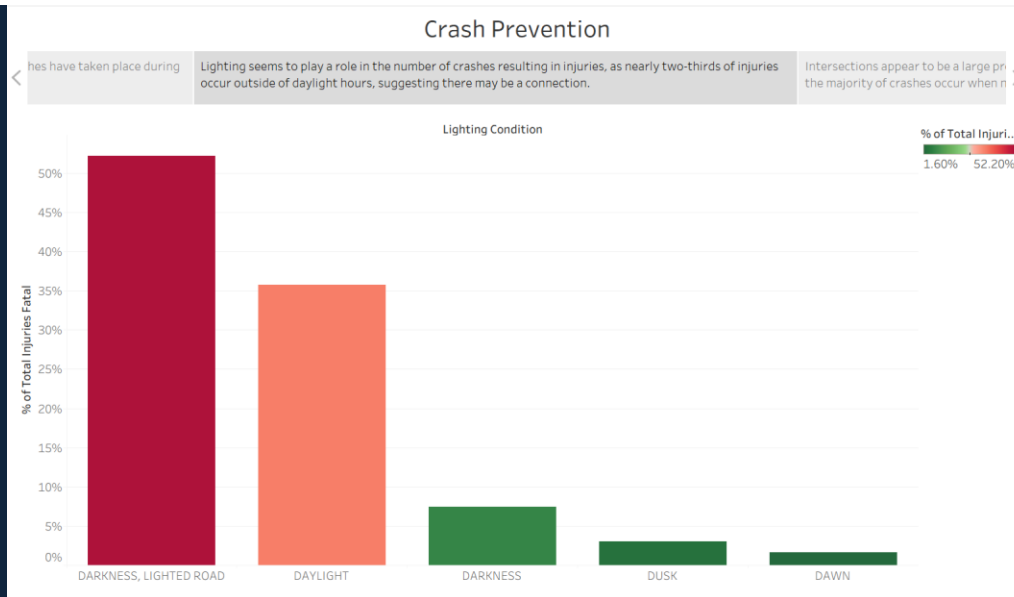
Recommendation 3: Emphasize Extreme Caution During Adverse Weather and Lighting

**Analysis of crash and injury data
by weather type, road condition,
and lighting condition**

Recommendation 3 Visuals + Analysis



- The data shows that many crashes occur during adverse weather conditions, as well as when roads are wet or covered in snow.
- In addition, over half of all injuries occurred outside of daylight hours despite more crashes occurring during broad daylight.
- If there were more advisories to be cautious during these times it could prevent some crashes, as drivers sometimes forget the risks of driving during adverse weather and at night.
- Another solution to this could be to have an informative website or videos that could remind people how to adjust their driving during unique weather and lighting conditions.





Recommendation 4: Emphasize Defensive Driving and Improve Driver Education

Analysis of crash data by first and second cause, driver action, and type

Recommendation 4 Visuals + Analysis

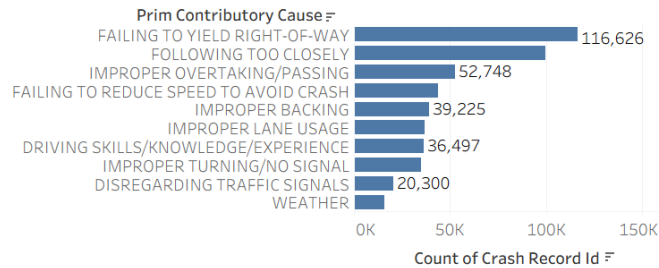
Crash Prevention

as, so that may be worth looking into further. In addition, the Airport, so that should certainly be evaluated.

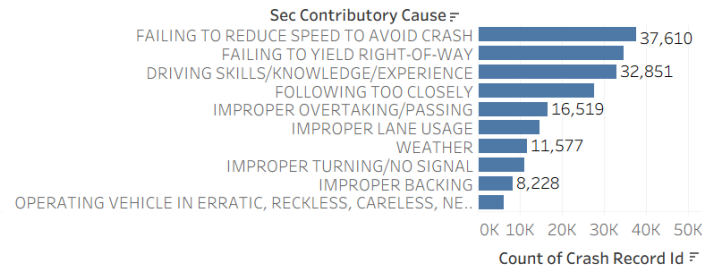
Unsurprisingly, most crashes occur because of preventable driver errors, with failure to yield by far the most common. Driving skills, knowledge, and experience has also caused an alarming amount of crashes.

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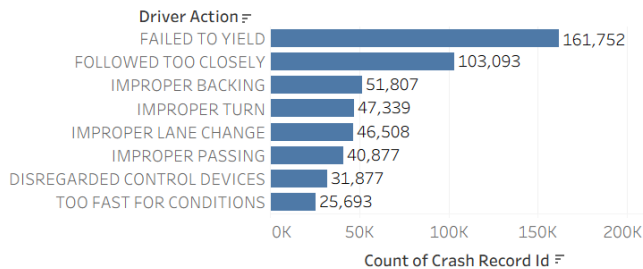
Crashes by 1st Cause



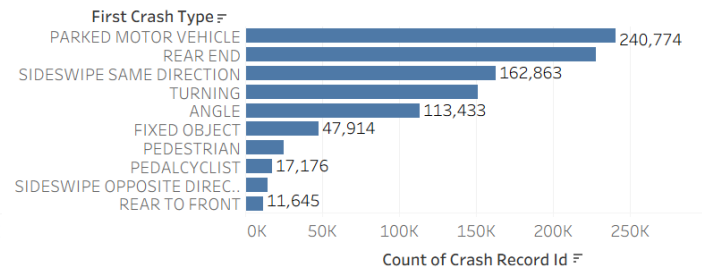
Crashes by 2nd Cause



Crashes by Driver Action



Crashes by Type



- The data indicates that a high number of crashes occur because of preventable driver errors, with many also happening in parking lots and due to lack of driving experience.
- It seems like improved driver education and possibly even steeper requirements to get a driver's license could help reduce some of these crash-causing errors.
- In addition, teaching and emphasizing defensive driving (assuming other drivers will do something wrong) and general caution on the road in advertisements and initial driving courses, some of these preventable crashes could likely be avoided.

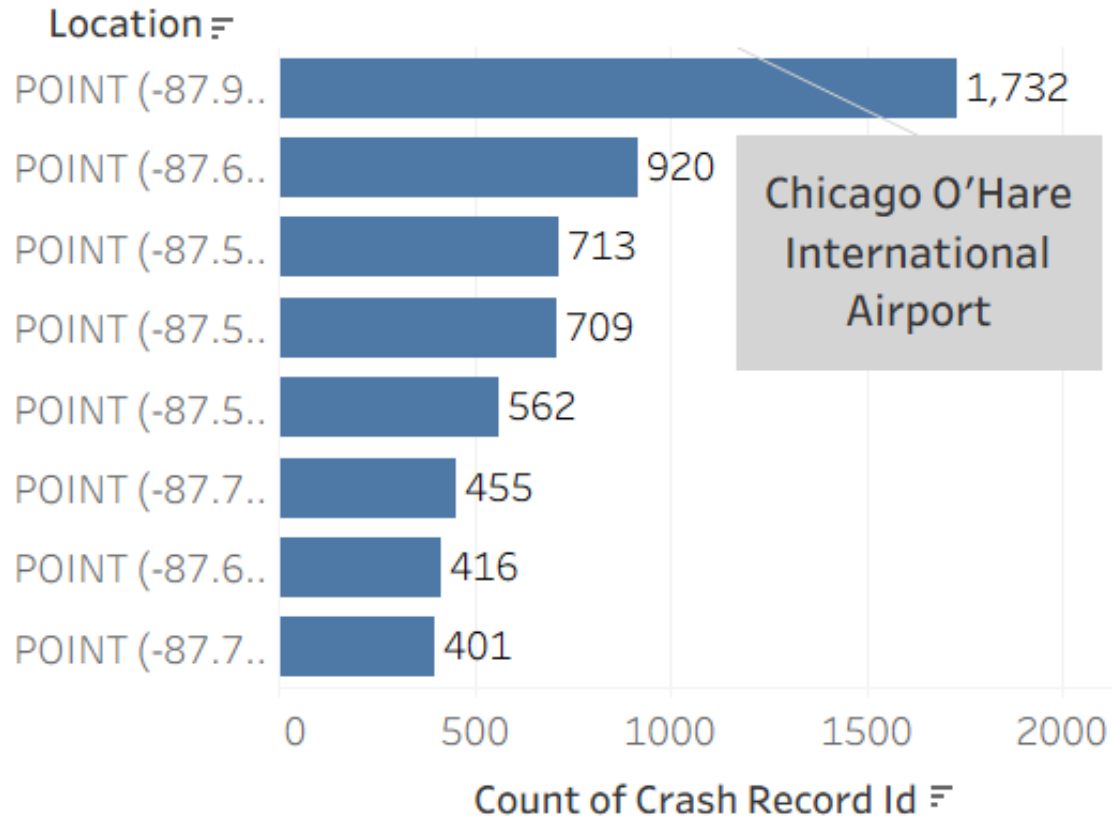


Recommendation 5: Increase Traffic Controls and Emphasize Caution at Airport

Analysis of crash data by coordinate

Recommendation 5 Visual + Analysis

Crashes by Coordinate



- There were a total of 1,732 crashes at Chicago's O'Hare International Airport, more than 800 more than any other coordinate.
- Even though airports have a high amount of security and law enforcement, there are a significant number of crashes that occur due to the high volume of drivers and traffic as well as people often being in a rush.
- To reduce this, a solution could be to increase the number of lanes at the airport terminals to spread cars out more and reduce traffic.
- Another solution could be to place personnel outside to help direct cars to their destination, which would reduce confusion, speed things up, and help cars have an easy entry and exit.

Final Brief

In conclusion, we hope that the solutions we presented can be implemented to help cut down on crashes in the city. By prioritizing these changes, Chicago can move towards safer driving conditions and significantly reduce the numbers of both crashes and injuries caused by them. The city will be able to have safer roads which both locals and tourists can feel comfortable on, which will make Chicago a much better and more enjoyable place to be overall.

